NCERT • GRADE 7 • MATHEMATICS

Constructions and Tilings

C06 • Ganita Prakash, Part II

Compass • Ruler • Geometric design

Perpendicular bisectors • Angles

Tangrams • Tiles • Symmetry

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. Illustrations are editable LaTeX drawings.

Part A • NCERT "Figure it Out" answers (textbook pp. 140–160)

How to use these geometric answers

Construct means draw with an unmarked ruler and compass; a finished drawing is part of the answer. The numbered sections below follow the *nine Figure it Out sets* in the chapter. For elaborate patterns, the method is explained and an editable example is shown: reproduce the particular outline shown in the NCERT textbook using the described arcs and repeated units.

Set 1 • Perpendicular bisectors (p. 140), Q1–Q3

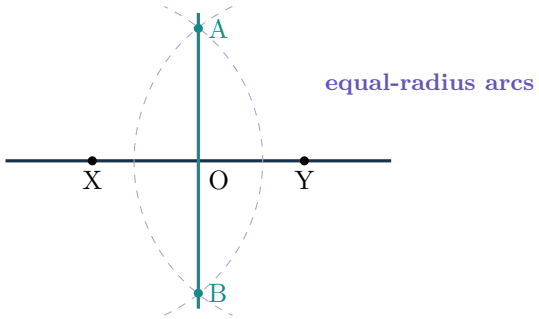
Q1. Same radius above and below XY? Not necessary. Each arc *pair* must use equal radii from X and Y, but the upper pair can have one radius and the lower pair another. Their intersection points are both equidistant from X and Y, so the line through them is the perpendicular bisector. Choose radii greater than half of XY for intersections.

Q2. Can both pairs be on the same side? Yes, choose two distinct suitable radii, and obtain two distinct points on the same side, each equidistant from X and Y. Their line is the perpendicular bisector.

Q3. Must each pair of arcs have equal radii? Yes, for this standard method: equal distances from X and Y guarantee the intersection lies on the perpendicular bisector. Arbitrary unequal radii do not give that guarantee.

Set 1, Q4 • Recreate the repeated design

Identify straight supporting lines, symmetry axes and intersection points in the textbook design. Use compass arcs of equal radius from symmetrical centres; repeat the same construction in each matching region and darken only the required final boundary. **Check:** equal parts must stay equal after copying; preserve the arcs faintly as construction lines.



Editable construction: AB bisects XY at right angles.

Set 2 • Rope-based constructions (p. 142), Q1–Q2

Q1. Why does the rope method bisect XY perpendicularly? A and B are formed from equal-length rope halves, so AX=AY and BX=BY. Both points lie on the perpendicular bisector, hence AB is that line.

Q2. Another rope method for a 90° angle: Mark X and Y at the same rope-distance either side of O on the given line. Pull a folded rope taut from X and Y to make point A away from the line. Join O to A; then $\angle AOX = 90°$.

Set 3 • Angle bisectors (pp. 144–145), Q1–Q3

Q1. Construct and bisect four angles: Example choices are $60°, 90°, 120°, 150°$. Their halves are $30°, 45°, 60°, 75°$, respectively. For each angle, cut equal distances on its arms, draw equal arcs from the two marks, and join the vertex to the arc intersection.

Q2. Eight-petal design (Fig. 6.5): Draw four full lines through a common centre, equally spaced by $45°$. Construct each successive bisector from a $90°$ angle, then add congruent curved petals symmetrically about each of eight sectors.

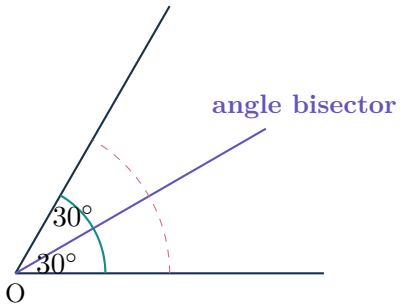
Q3. Intersecting arcs on the opposite side: The *line* through the vertex and the opposite intersection remains aligned with the angle bisector. But if point C is on the exterior side, the *ray* OC itself points outside the given angle and is not its interior bisector. Extend OC through O to get the interior ray.

Set 3, Q4–Q6 • Other angle constructions and designs

Q4. Repeated bisection gives $45°, 30°, 15°, 22.5°$ from constructible starting angles. **A general $65.5°$ angle cannot be obtained by these simple ruler-and-compass bisections**; a protractor may be used if measurement is allowed.

Q5. Rope method: mark equal distances OA=OB, then use two equal rope lengths from A and B to locate C; join O to C. SSS congruence gives two equal angles.

Q6. Petals fitting a square: Draw the diagonals and midlines as axes; use midpoints, symmetries, and equal-radius arcs to copy each petal. Increase each radius symmetrically until the petals just touch the intended boundary without crossing it.



Example: a $60°$ angle is divided into two $30°$ angles.

Set 4 • Copy angles (p. 146), Q1–Q2

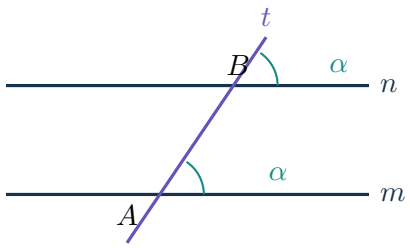
Q1. Copy four differently oriented angles. (1) Draw an arc centred at the first vertex, cutting its arms at A and B. (2) Draw an arc of the same radius at a new vertex. (3) Transfer the chord length AB onto the new arc using the compass. (4) Join the new vertex to the marked point. Repeat for four originals. This is a congruent-triangle construction, not estimation by eye.

Q2. Construct Fig. 6.6. Identify its repeating bent-line unit. Make one unit accurately, copy its angle using the chord-transfer method, and copy its arm lengths using a compass. Alternate the shown orientations without altering lengths or angles.

Set 5 • Parallel lines (p. 148), Q1–Q2

Q1. Construct four pairs of parallel lines: Draw a transversal of a chosen original line; at another point on the transversal copy its corresponding angle. The second line is parallel to the first. Repeat with horizontal, vertical and two sloping examples.

Q2. Recreate the line design: Follow the NCERT figure's marked points; build repeated segments through copied angles and equal lengths. Use the **corresponding-angle rule** to create each parallel element, then trace its boundary. Verify the intended crossings rather than simply drawing freehand.



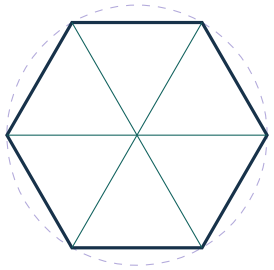
Set 6 • Arch designs (p. 150), Q1–Q2

Q1. Construct pointed arches based on Fig. 6.11: Begin with the equal support segments, locate their midpoints, then draw two congruent arcs so they meet at an apex above the centre. Vary arc radius while keeping the left/right construction symmetric.

Q2. Create your own arch designs: For example, copy one pointed arch repeatedly side-by-side, or add a smaller nested pair of arcs. Label the central line of symmetry and the two equal radii used.

Set 7 • Hexagons, patterns and perpendiculars (pp. 154–155), Q1

(a) Inflexed arc: use adjoining circular arcs meeting with a change in curvature; mark the centres using equal support lengths.
(b) Compass-only figure: transfer the same circle radius repeatedly to establish congruent points and arcs, then darken the required outline.
(c) Regular hexagon in a circle: draw a circle; keep its radius unchanged and step six equal chords around its circumference; join adjacent points.
(d) Six-circle rosette: mark six equally spaced centres on a circle; draw six congruent circles centred on them, each with the appropriate tangent radius.
(e) Intricate repeating hexagonal motif: first make a regular hexagon and its three long diagonals; bisect repeated edges or angles and copy the same triangular cell motif sixfold.
Note: refer to the textbook figures for the exact final tracings in (a), (b), (d) and (e); the steps identify the repeatable construction and preserve its symmetry.



Six equal 60° sectors make a hexagon

Set 7, Q2–Q4 • Optical illusion, star and exterior-point perpendicular

Q2. Optical illusion: Three sharp corner arrangements and three incomplete circular discs can make us *perceive* contours of a triangle that are not actually drawn (subjective/illusory contours). Reproduce it by copying three symmetric repeated units around the centre.

Q3. Star inside a hexagon: Construct the outer regular hexagon, mark its central 60° directions, and use equal marked distances on corresponding sides to build the symmetric six-pointed outline. Check each matching angle before darkening the star.

Q4. Perpendicular through P outside ℓ: With centre P draw an arc cutting ℓ at X and Y. Since PX=PY, P lies on XY's perpendicular bisector. Construct the perpendicular bisector of XY; it passes through P and meets ℓ at 90°.

Set 8 • Tangram challenges (p. 156)

A tangram has **seven pieces**: five triangles, one square and one parallelogram. To form each silhouette pictured in the exercise, (1) trace its outer boundary, (2) place the largest two triangles first, (3) fit the medium triangle and quadrilateral pieces in remaining corners, then (4) fill gaps with the two smallest triangles. Use each of the seven pieces exactly once, permitting rotations and flips where the physical pieces allow. The textbook silhouettes are open construction tasks; more than one arrangement may work.

Set 9 • Tiling tests (p. 160), Q1–Q2

Q1: YES. The L-shaped region contains 12 squares, and each L-triomino covers 3 squares. An exact arrangement of four L-triominoes fills the region without gaps or overlaps (see the grid below).

Q2: NO. An 8×8 square board has two opposite corners removed. The remaining 62 squares include **unequal counts of the two chessboard colours**: both removed corners are the same colour. Each 2×1 domino always covers one square of each colour, so the region cannot be completely tiled.

1	1		
1	2		
3	2	2	4
3	3	4	4

Four groups, each of **three** connected squares

$4 \times 3 = 12$ **covered squares**

One valid tiling for textbook p. 160, Q1; equal numbers indicate one L-triomino.

Why counting only area is not enough

62 is even, but that does *not* prove a shape is tileable by dominoes. For the two-corner problem, the **black/white colouring invariant** proves that it is impossible.

Part B • Essential constructions: quick methods

Four methods to memorise

Perpendicular bisector: intersecting equal-radius arc pairs from the two endpoints.
Angle bisector: equal steps on both arms, then equal-radius intersecting arcs.
Copy an angle: duplicate a radius and transfer a chord length.
Draw a parallel: copy a corresponding angle across a transversal.

Part C • Why the constructions work

Logic behind the drawings

Perpendicular bisector: a point equidistant from X and Y is on XY's perpendicular bisector. Two such points determine the line.
Bisected angle: the two small triangles share a side and have matching other sides (SSS), so their angles at the main vertex are equal.
Regular hexagon: one full turn is 360°, and $360°/6 = 60°$; each sector is an equilateral triangle.
Tiling: a complete cover requires *no gaps and no overlaps*, not merely the correct total area.

Part D • Extra practice (not textbook questions)

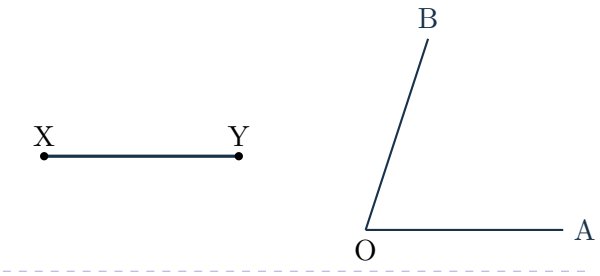
Check your understanding

1. A 100° angle is bisected. Find each angle. 50°.
2. How many 60° sectors are there around a point? **Six.**
3. Can a 5×7 grid be covered with 2×1 dominoes? **No**: 35 is odd.
4. Can a 4×7 grid be tiled with dominoes? **Yes**: use 14 vertical dominoes in two-row bands.
5. What is the interior angle of a regular hexagon? 120°.
6. Which instrument transfers exact lengths? **Compass.**

Part E • Construction sketchbook (try in your notebook)

Three mini practical challenges

A. Using a compass, construct the perpendicular bisector of XY below.
B. Bisect the angle $\angle AOB$ without a protractor.
C. Starting from an equilateral triangle, construct a regular hexagon and explain why six such triangles fit around one centre.



Keep your construction arcs visible and label every important point.

Reference: NCERT *Ganita Prakash*, Grade 7, Part II, Chapter 6, pp. 136–163.